HY Analyst: from method to code

How the high yield mispricing method works, and where each part lives in `hy/`

FundamentalQuantmental

21 October 2026

The method in one page

The question. Which high yield bonds pay more spread than their risk justifies?

The answer in four moves.

1. Measure the issuer today: earnings, cash, debt, and risk flags (Section 2).
2. Project the issuer forward under named scenarios. Let the macro forecast set how likely each scenario is (Section 3).
3. Turn the projections into a probability of default and a loss given default. Their product, scaled by the market price of risk, is the *anchor spread*. A second model explains what the anchor misses (Sections 4 and 5).
4. Compare the market spread with the fair spread. A bond is cheap only when it sits outside a band that measures the model's own error (Section 5). Seven signals and an event risk model turn that gap into positions (Sections 6 and 7).

The base. Gatto, *The Credit Investor's Handbook* (Wiley, 2023), gives the analyst process. The book has no statistical models. This system keeps the book's process as the feature set and adds the models around it. `docs/methodology.md` is the full specification; its section numbers match the module docstrings.

The rule that protects every result. Every input carries two dates: `as_of` (the period it describes) and `available_at` (the first moment the market could read it). A model reads only fields available before the decision time. A test enforces this.

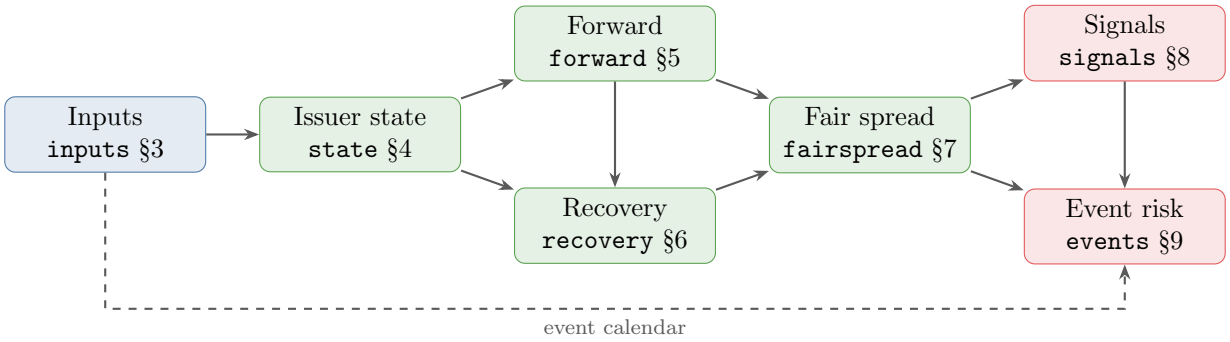


Figure 1: One spine, two consumers. Blue: data. Green: models. Red: what the desk uses. Every table written by the pipeline follows the schemas in `outputs` (§10).

What this means for you

- **The code runs end to end on synthetic data today.** From the repository root run `PYTHONPATH=. python3 scripts/demo_pipeline.py`.

It takes one synthetic universe through Sections 5 to 8 and prints a force rank.

- **The code has not seen real data yet.** The real-data step is to load filings, TRACE prices and macro series into the `inputs` records. Your `marketdb` has TRACE but not issuer financials with release dates.
- **Each claim in the method is now a test.** Section 8 lists them. A green test means the formula does what the method says. It does not yet mean the signal makes money.

1 Inputs and the look-ahead guard (§3, `hy/inputs.py`)

Book. The analyst reads the 10-K, the indenture, and the market (Chapters 3, 5, 12).

Code. Each record type (`IssuerFinancials`, `Tranche`, `Event`) inherits `Stamped`, which holds `key`, `as_of` and `available_at`. `PointInTimeStore.view(t)` returns the latest version of each record that the market could see at time t. A restatement is a new version of the same key with a later `available_at`; it replaces the original only after its own release.

Intuition: a backtest that reads a restated number before the restatement date finds alpha that never existed.

2 Issuer state (§4, `hy/state.py`)

Book. Steps 2 and 3: qualitative risks, profitability, cash flow, capital structure (Chapters 6–9).

Adjusted EBITDA. Add back only items that do not recur:

$$EBITDA_t^{adj} = EBITDA_t^{rep} + \textstyle\sum_i a_i\, NR_{i,t}, \qquad a_i \in \{0,1\}.$$

Intuition: a "one-off" charge that comes every year is an operating cost.

Free cash flow. The code charges the cash part of each add-back:

$$FCF_t = EBITDA_t^{adj} - CashNR_t - Capex_t - CashInterest_t - CashTax_t - \Delta WC_t.$$

Intuition: an add-back for a cash restructuring charge raises earnings but not cash.

`methodology.md` §4.2 adopted this on 21 October 2026.

Leverage through the tranche. Senior claims at face, the tranche under analysis at market:

$$Lev_k^{face} = \frac{\sum_{rank_j \le rank_k} F_j}{EBITDA^{adj}}, \qquad Lev_k^{mkt} = \frac{\sum_{rank_j < rank_k} F_j + P_k F_k}{EBITDA^{adj}}.$$

Intuition: Lev_k^{mkt} is the creation multiple: the company value at which a buyer of this bond owns the business.

Covenant headroom. $H_t = 1 - Lev_t^{cov}/Lev_t^{max}$. At origination lenders set $Lev^{max} = D/((1-c)\,EBITDA^{mgmt})$; a 15% cushion on a 4.0x plan gives 4.7x (p. 356).

Rating gap. An ordinal model predicts the rating from scale, leverage, coverage, margin and sector. Fitted rating minus agency rating, in notches, is the gap.

Intuition: ratings lag prices (p. 274), so a bond rated above its fundamentals is a downgrade candidate.

3 Forward estimates (§5, `hy/forward.py`)

Book. Build management, base and downside cases; assign probabilities by judgement (Chapter 10).

Scenario engine. For scenario s and quarter h:

$$Rev_h^s = Rev_{h-1}^s(1+g_h^s), \qquad EBITDA_h^s = Rev_h^s(1-v_h^s-a_h^s),$$
$$FCF_h^s = EBITDA_h^s - Capex_h^s - i_h(m)D_{h-1}^s - Tax_h^s - \Delta WC_h^s.$$

Negative FCF draws the revolver. Interest uses last quarter's debt, so the engine needs no circular solver.

Intuition: liquidity and leverage come out of one recursion, so they cannot contradict each other.

Macro sets the weights. A pooled logistic model gives the chance of a large EBITDA fall in the next four quarters:

$$P(EBITDA_{t+4} < (1-\delta_i)EBITDA_t \mid m_t, sector_i) = \sigma(\theta_0 + \theta^\top m_t + \theta_{sector}).$$

That probability is the weight of the moderate downside. The severe case takes a fixed fraction of it, and the base case takes the rest.
Intuition: in a late cycle every issuer's downside weight rises before any issuer misses.

Forward features. Weighted mean and spread of leverage at 4 and 8 quarters, the peak downside leverage, the minimum downside liquidity, and the probabilities of covenant breach, cash shortfall and failed refinance. The book's own check (p. 311) is the flag $w_{down} > 0.5$: then the downside should be the base case.

4 Valuation and recovery (§6, `hy/recovery.py`)

Book. Four valuation methods (Chapter 11), priority of claims (Chapter 9), and documents that move claims ahead of yours (Chapter 18).

Value as a distribution. $\log V \sim N(\mu_V, \sigma_V^2)$, $\mu_V = \log V_{mid}$, $\sigma_V = (\log V_{high} - \log V_{low})/(2 \times 1.28)$.
Intuition: the low and high ends of the football field are read as the 10th and 90th percentiles.

Waterfall. Tranche k owns the slice of value between S_k (all claims ahead) and $S_k + F_k$:

$$Pay_k(V) = \min\big(F_k, \max(0, V - S_k)\big), \qquad R_k = E[Pay_k(V)]/F_k.$$

This is a call spread on firm value. With lognormal V it has a closed form, so the code needs no simulation; a Monte Carlo check confirms it.

Liability management. $R_k^{adj} = (1 - \pi_{LME})R_k + \pi_{LME} R_k^{LME}$. The probability π_{LME} rises with loose baskets, a bare-majority amendment vote, sponsor history, and closeness to distress.
Intuition: after the Serta-style uptier, PU Corp recovery fell from 70–82.5% to 24.8–49.9% (pp. 503–504).

5 Fair spread and mispricing (§7, `hy/fairspread.py`)

Book. Risk is probability of default times loss given default (p. 119). The book compares bonds by judgement; it has no spread model.

Anchor. $\lambda_i = -\log(1 - PD_i^{12m})$, $s_i^{EL} = \lambda_i LGD_i^{adj}$, $s_{i,t}^{anchor} = \kappa_t \, s_i^{EL}$. The PD model is calibrated so that its mean by rating matches the agency default table.
Intuition: κ_t is the market price of credit risk: high after a sell-off, low in a frothy market.

Second stage. $\log s_{i,t} = \log s_{i,t}^{anchor} + f_t(x_{i,t}) + \eta_{sector,t} + \varepsilon_{i,t}$. The model f_t is a gradient-boosted tree with monotone constraints: the spread must rise with leverage and fall with coverage.
Intuition: the constraints stop the model from learning a wrong sign, so a large residual is never a sign error.

The band. $z = \varepsilon/\hat{\sigma}(x)$. Split conformal prediction, run separately in each rating and liquidity cell, gives a threshold $q_{1-\alpha}$. A bond is cheap when $z > q_{1-\alpha}$. Adaptive conformal inference moves α_t so that coverage stays on target as spreads drift.
Intuition: illiquid CCC bonds are noisy; the band is wider there, so the model does not call noise "cheap".

Convergence. An AR(1) on each residual gives the half-life $\log 2/(-\log \rho)$ and the expected return $carry_h + D^s(1 - \rho^h)m$.

6 Signals (§8, `hy/signals.py`)

Signal	What it buys	Main failure
S1 residual	Spread above the conformal band	Fact the model misses
S2 momentum	Fundamentals improved more than the spread	Non-cash improvement
S3 revision	Forward leverage falls; analyst below consensus	Analyst bias
S4 capital structure	Tranche gap above the recovery gap	Both legs go to recovery
S5 documents	In-the-money change-of-control put; protected tranche	Fall-away clause
S6 ratings	Positive rating gap, before the upgrade	Already priced
S7 sector	Sector cheap to its macro forecast	Stale forecast

The composite weights each signal by its trailing rank IC in the current cycle phase; a signal with negative IC gets zero weight. The composite is made neutral to index beta and duration. The force rank sorts the whole universe, not each sector (p. 369). No position enters without `replaces_position`.

7 Event risk (§9, `hy/events.py`)

A dated event (earnings, covenant test, maturity, rating review, LME) has a set of outcomes. A macro forecast is four to six named paths. For each outcome o and path m the code changes the features and re-evaluates the fair spread:

$$\Delta P_i(o,m) \approx -D_i \Delta y(m) - D_i^s \left[\Delta s^{sys}(m) + \hat{s}_i(x_i(o,m)) - \hat{s}_i(x_i)\right] + carry_h - \mathbb{1}\{default\}(P_i - 100R_i^{adj}).$$

Intuition: the model that says a bond is cheap also says how much it loses on a miss.

Why yields need a split. $\Delta s^{sys} = \beta_g \Delta y^{growth} + \beta_p \Delta y^{policy}$ with $\beta_g < 0$, $\beta_p > 0$. Yields up on strong growth tighten spreads; yields up on Fed tightening widen them. So a high yield bond has a short effective rate duration in a growth move and a long one in a policy move.

Pair trades. Size pairs by jump-to-default, not DV01: $N_j/N_k = (P_k - 100R_k)/(P_j - 100R_j)$. In a filing both legs converge to recovery (p. 530).

8 What the tests prove

Run `python3 -m pytest` from the repository root. On 21 October 2026, 118 tests passed in about 5 seconds.

Module	Section	Tests	Key claims tested
`inputs`	3	4	A filing is hidden before release; a restatement replaces the original only after its own release
`state`	4	21	15% cushion on 4.0x gives 4.7x; lease adjustment; ordinal model recovers ratings
`forward`	5	16	Cash and debt reconcile each quarter; higher rates raise leverage; downside $\geq$ base; weights sum to 1 and rise with macro stress
`recovery`	6	18	Closed-form recovery matches Monte Carlo within 1%; senior $\geq$ junior; recovery falls as senior claims grow
`fairspread`	7	13	Spread rises in leverage and falls in coverage; conformal coverage in every cell; PD by rating matches the table; HCA spread per turn (172 and 35 bp)
`signals`	8	19	S1 is zero inside the band; composite neutral to beta and duration; no position without `replaces_position`
`events`	9	14	Growth-driven yield rise tightens spreads, policy-driven widens; DV01-neutral pair is not default-neutral
`outputs`	10	5	Schema check catches a row with `available_at` after `date`

End-to-end check. `scripts/demo_pipeline.py` builds 300 synthetic issuers over 12 dates. It plants a 0.45 log-spread cheapness in 15 bonds on the last date. The pipeline finds 11 of the 15 in the top 15 of the force rank. It raises 15 cheap flags; 4 are false, against about 7 that the band permits at $\alpha = 0.05$.
Intuition: the wiring works. This is not evidence of alpha, because the synthetic data follow the same structure the models assume.

Limit of one test. The coverage test passes on its fixed seed and fails on 2 of 12 other seeds. That is the expected false-alarm rate for six cells at two standard errors. On real data, judge coverage over many dates, not one run.

9 Open decisions

1. **S2 sign (resolved 21 October 2026).** `methodology.md` §8 had $S2 = \Delta \log \hat{s} - \Delta \log s$, which gives a short when fundamentals improve. It now reads $S2 = \Delta \log s - \Delta \log \hat{s}$, which matches its own text and the code. `tests/test_signals.py` pins the sign.

2. **Downside weights (resolved 21 October 2026).** The moderate downside takes $(1 - f)p$ and the severe case fp, with $f = 0.25$, so together they equal the probability of a large decline. `methodology.md` §5.2 now says this. The value of f still needs a source.

3. **FCF start point (resolved 21 October 2026).** FCF starts from adjusted EBITDA and subtracts the cash part of the add-backs. `methodology.md` §4.2 now says this. On real data, check that it tracks operating cash flow minus capex.

4. **Placeholder numbers.** Only the speculative-grade default rates (3.60% and 14.16%) come from the book. The PD table for BB, B and CCC, the constant-hazard 5-year PD (16.7% against the book's 14.16%), the rating-migration curve, the outcome splits in §9, the severe-downside fraction, and the LME coefficients γ use stated placeholders. Each needs a calibration on real data before any signal is trusted.

5. **Data.** The pipeline needs issuer filings with SEC acceptance times, consensus vintages, and indenture fields. None of these is in the repository yet.